

Practice sheet 2 lab:

DSA

Problem 1: Array Traversal

```
import java.util.Scanner;
```

```
public class ArrayTraversal {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        for(int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        System.out.println("Array Elements:");
```

```
        for(int i = 0; i < n; i++) {
```

```
            System.out.print(arr[i] + " ");
```

```
        }
```

```
}  
}
```

Problem 2: Insert an Element at a Given Position

```
import java.util.Scanner;  
  
public class InsertElement {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
  
        int[] arr = new int[n + 1];  
  
        for(int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }
```

```
int element = sc.nextInt();
int position = sc.nextInt();

for(int i = n; i >= position; i--) {
    arr[i] = arr[i - 1];
}

arr[position - 1] = element;

System.out.println("Updated Array:");

for(int i = 0; i <= n; i++) {
    System.out.print(arr[i] + " ");
}

}

}
```

Problem 3: Delete an Element from an Array

```
import java.util.Scanner;
```

```
public class DeleteElement {  
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        for(int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }
```

```
        int position = sc.nextInt();
```

```
        for(int i = position - 1; i < n - 1; i++) {  
            arr[i] = arr[i + 1];  
        }
```

```
        System.out.println("Array After Deletion:");

        for(int i = 0; i < n - 1; i++) {
            System.out.print(arr[i] + " ");
        }

    }
}
```

Problem 4: Find Largest and Smallest Element

```
import java.util.Scanner;

public class LargestSmallest {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
```

```
int[] arr = new int[n];
```

```
for(int i = 0; i < n; i++) {  
    arr[i] = sc.nextInt();  
}
```

```
int max = arr[0];
```

```
int min = arr[0];
```

```
for(int i = 1; i < n; i++) {
```

```
    if(arr[i] > max) {  
        max = arr[i];  
    }
```

```
    if(arr[i] < min) {  
        min = arr[i];  
    }
```

```
}
```

```
        System.out.println("Maximum = " + max);
        System.out.println("Minimum = " + min);

    }
}
```

Problem 5: Count Even and Odd Numbers

```
import java.util.Scanner;

public class EvenOddCount {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];
```

```
for(int i = 0; i < n; i++) {  
    arr[i] = sc.nextInt();  
}
```

```
int even = 0;
```

```
int odd = 0;
```

```
for(int i = 0; i < n; i++) {
```

```
    if(arr[i] % 2 == 0) {
```

```
        even++;
```

```
    } else {
```

```
        odd++;
```

```
    }
```

```
}
```

```
System.out.println("Even = " + even);
```

```
System.out.println("Odd = " + odd);
```


}

}

End of sheet 2

Thank you